# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 8 — MATHS FULL SYLLABUS PAPER SET 08

**Full Syllabus (Chapters 1–14)** | RBSE Pattern | Session 2026–27

**Time: 3 Hours 15 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 80**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

## SECTION — A (Objective Type Questions) 20 × 1 = 20

*Choose the correct option and write it in your answer book.*

**1.** Which of the following is a perfect square? 1

**(a)** 81 **(b)** 120 **(c)** 82 **(d)** 90

**2.** The value of $9^3$ is: 1

**(a)** 729 **(b)** 720 **(c)** 738 **(d)** 243

**3.** $7^3 \times 7^3$ equals: 1

**(a)** $7^0$ **(b)** $7^9$ **(c)** $7^6$ **(d)** $49^6$

**4.** The decimal number 25 written in base 2 is: 1

**(a)** 11001 **(b)** 11010 **(c)** 11000 **(d)** 11011

**5.** A parallelogram with equal diagonals is a: 1

**(a)** trapezium **(b)** rectangle **(c)** kite **(d)** rhombus

**6.** In a parallelogram ABCD, angle A is 110°. Angle C is: 1

**(a)** 110° **(b)** 250° **(c)** 70° **(d)** 90°

**7.** The smallest digit that replaces the blank in 47_5 so that the number is divisible by 3 is: 1

**(a)** 3 **(b)** 4 **(c)** 2 **(d)** 7

**8.** $(a - b)^2$ is always: 1

**(a)** negative **(b)** odd **(c)** zero **(d)** non-negative

**9.** If 5 notebooks cost ₹ 120, then 8 notebooks cost: 1

**(a)** ₹ 240 **(b)** ₹ 192 **(c)** ₹ 160 **(d)** ₹ 200

**10.** If x and y are inversely proportional and x doubles, y: 1

**(a)** doubles **(b)** becomes half **(c)** stays same **(d)** becomes zero

**11.** A quantity of 450 is increased by 20%. The new value is: 1

**(a)** 540 **(b)** 630 **(c)** 360 **(d)** 90

**12.** $\frac{5}{8}$ expressed as a percentage is: 1

**(a)** 62.5% **(b)** 125% **(c)** 57.5% **(d)** 67.5%

**13.** Is a triangle with sides 14, 48 and 50 a right triangle? 1

**(a)** Only if it is isosceles **(b)** Cannot say

**(c)** No **(d)** Yes

**14.** The legs of a right triangle are 7 cm and 24 cm. Its hypotenuse is: 1

**(a)** 26 cm **(b)** 31 cm **(c)** 25 cm **(d)** 24 cm

**15.** In a pie chart, a sector representing 10% of the data has a central angle of: 1

**(a)** 10° **(b)** 20° **(c)** 36° **(d)** 46°

**16.** A solid has 6 faces and 8 vertices. Using $F + V - E = 2$, the number of edges is: 1

**(a)** 16 **(b)** 12 **(c)** 14 **(d)** 10

**17.** In a line graph, a steep upward line segment shows: 1

**(a)** no change **(b)** a rapid increase **(c)** a decrease **(d)** an error

**18.** In a calendar, the dates in a $2 \times 2$ block starting at 5 are 5, 6, 12, 13. Their sum is: 1

**(a)** 34 **(b)** 36 **(c)** 38 **(d)** 28

**19.** The area of a triangle with base 20 cm and height 7 cm is: 1

**(a)** 140 cm² **(b)** 27 cm² **(c)** 70 cm² **(d)** 90 cm²

**20.** A rectangle has area 150 cm² and length 15 cm. Its breadth is: 1

**(a)** 12 cm **(b)** 135 cm **(c)** 10 cm **(d)** 9 cm

## SECTION — B (Very Short Answer Type) 8 × 2 = 16

*Answer in about 10–15 words each.*

**21.** Why is any number raised to the power 0 equal to 1 (for non-zero numbers)? 2

**22.** In base 60 (Mesopotamia), how many minutes make an hour? Why is that a relic of base 60? 2

**23.** Solve the cryptarithm: $A + A = 10$. Find $A$. 2

**24.** What does a map scale 1 : 50000 mean? 2

**25.** Name the hypotenuse in a right triangle PQR right-angled at Q. 2

**26.** State the difference between direct and inverse proportion. 2

**27.** Marks of 5 students: 40, 45, 50, 55, 60. Find the mean and median. 2

**28.** Two parallelograms on the same base have heights 5 cm and 8 cm. Find the ratio of their areas. 2

## SECTION — C (Short Answer Type) 8 × 3 = 24

*Answer in about 30–40 words each.*

**29.** Find the square of 78 using $(a + b)^2$ with $a = 80$, $b = -2$. 3

**30.** A trapezium has parallel sides 10 cm and 6 cm and height 4 cm. Find its area. 3

**31.** Expand $(3m - 2n)^2$ and find its value when $m = 2$, $n = 1$. 3

**32.** Find the compound interest on ₹ 8000 at 5% per annum for 3 years. 3

**33.** Describe the Koch snowflake construction in three steps. 3

**34.** Two numbers add to 18. Find the numbers that give the largest product and the product. 3

**35.** A rectangular park measures 8 m by 15 m. A boy walks along the diagonal instead of two sides. How much distance does he save? 3

OR

A ladder of length 25 m rests against a wall with its foot 7 m from the wall. How high up the wall does it reach? 3

**36.** Find the area of a trapezium with parallel sides 9 cm and 15 cm and distance between them 10 cm. Also find the area of an equal-height rectangle with width 15 cm. 3

OR

A rectangular plot is 20 m by 13 m. Find its area and the cost of fencing at ₹ 30 per metre. 3

## SECTION — D (Long Answer Type) — 4 × 5 = 20

*Answer in about 80–100 words each.*

**37.** (i) Prove $a^m \times a^{-m} = 1$. 5
(ii) Verify for $a = 3,\ m = 2$.

OR

(i) Explain how to compare two numbers written in scientific notation. 5
(ii) Compare $4.5 \times 10^7$ and $9.1 \times 10^6$.

**38.** **Case study:** A farmer borrows ₹ 50000 at 12% simple interest for 3 years. 5
(i) Find the interest.
(ii) Total to be repaid.
(iii) Monthly instalment if repaid equally over 36 months.

OR

(i) In a class, 60% of 40 students are girls. Of these, 25% have bicycles. How many girls have bicycles? 5
(ii) What percent of the class are they?

**39.** (i) Show that the sum of deviations of values from their mean is zero for 4, 6, 8, 10, 12. 5
(ii) Explain the balancing idea.

OR

(i) Find $x$ if the median of 3, 5, $x$, 9, 11 is 8. 5
(ii) Find the mean.

**40.** (i) Find the area of a parallelogram with base 24 cm and height 15 cm. 5
(ii) If its slant side is 17 cm, find the area of the triangle cut off by the height.

OR

(i) A square and a rectangle have equal perimeter 48 cm; the rectangle's length is 16 cm. Compare their areas. 5
(ii) What do you conclude?

## END OF QUESTION PAPER — Total: 80 Marks

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